

EMSCAD

Complete Exploratory Data Analysis

for Fraudulent Job Posting Detection

Dataset: Employment Scam Aegean Dataset
17,880 Job Postings | 18 Columns | 866 Fraudulent Posts

Prepared for the SAGE-FJD Research Project
Leakage-Safe Multimodal Fraudulent Job Detection

Report generated: 25 August 2026

Executive Summary

Purpose

This report presents a complete exploratory data analysis of EMSCAD before developing the SAGE-FJD fraudulent job posting detection model. The analysis examines data quality, target imbalance, textual behaviour, structured metadata, missingness, suspicious language, repeated templates and campaign-level leakage risk.

Main dataset characteristics

EMSCAD contains 17,880 job postings and 18 columns. Only 866 postings are fraudulent, producing a fraud prevalence of 4.84%. Several fields contain substantial missing values, and many postings reuse identical or nearly identical text templates.

Most important EDA finding

The most important finding is that repeated templates can appear across many rows. A normal random train-test split could place copies from the same campaign into different partitions and produce misleadingly high model performance.

Resulting research decision

The modelling protocol must use campaign-group-isolated training, validation, test and five-fold out-of-fold splits. The final model should combine lexical, structured metadata, semantic and campaign-level evidence.

EDA Methodology

Dataset inspection

The analysis checks dataset shape, data types, missing values, target distribution, exact duplicates, repeated content, unique-value cardinality and class-specific descriptive statistics.

Fraud association analysis

Fraud rates are calculated for binary and categorical features, missingness patterns, country, industry, function, salary availability and suspicious text signals.

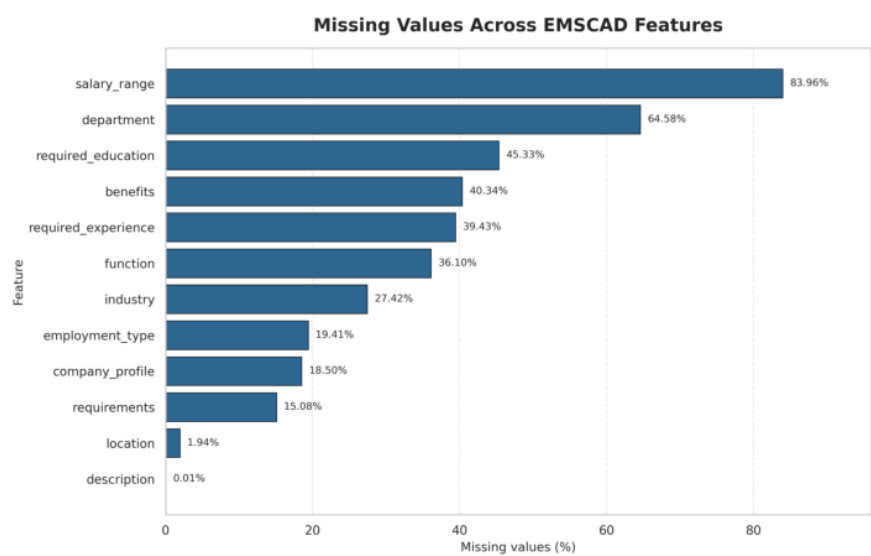
Text and template analysis

Text lengths, associated terms and normalized template repetition are studied separately for title, company profile, description, requirements and benefits.

Statistical caution

Observed percentages for rare groups must be interpreted carefully. Association does not establish causation, and EDA results must not be used as direct fraud-rejection rules.

Figure 1: Missing-Value Distribution



What the graph shows:

This graph presents the number and percentage of missing values in every EMSCAD feature. Salary range has the highest missingness at approximately 83.96%, followed by department at 64.58%, required education at 45.33%, benefits at 40.34%, required experience at 39.43%, function at 36.10%, and industry at 27.42%. Location and description are comparatively complete.

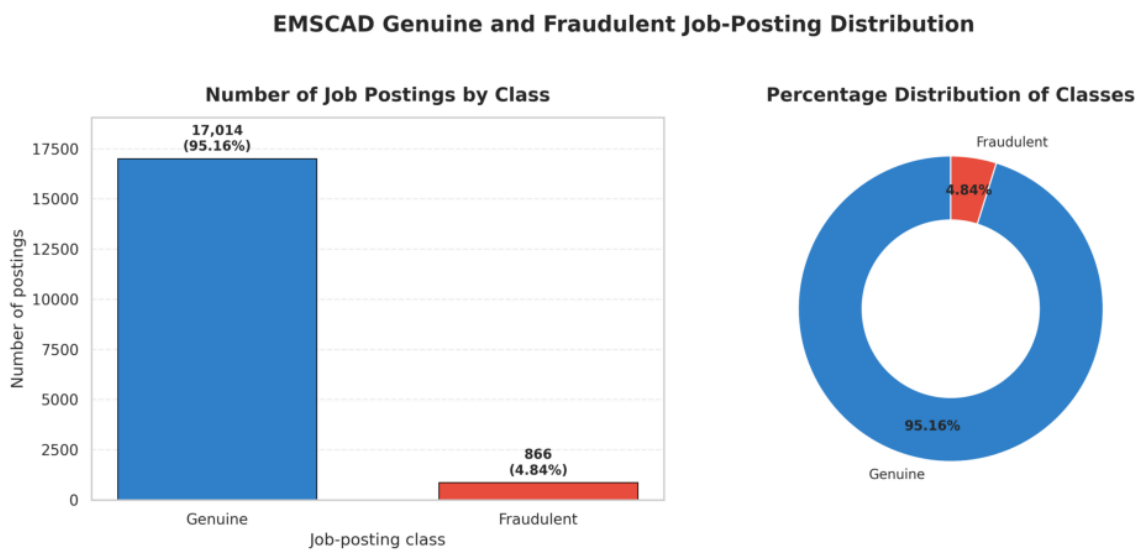
Research interpretation:

Missing data is not distributed equally across the dataset. Removing every row containing a missing value would discard a large portion of EMSCAD and could remove useful fraud patterns. Therefore, missingness must be treated as information rather than only as a data-quality problem.

Modelling implication:

Text fields should use an explicit missing-section token. Categorical values should use a Missing category, while separate missing-indicator features should be created for the metadata model.

Figure 2: Genuine and Fraudulent Class Distribution

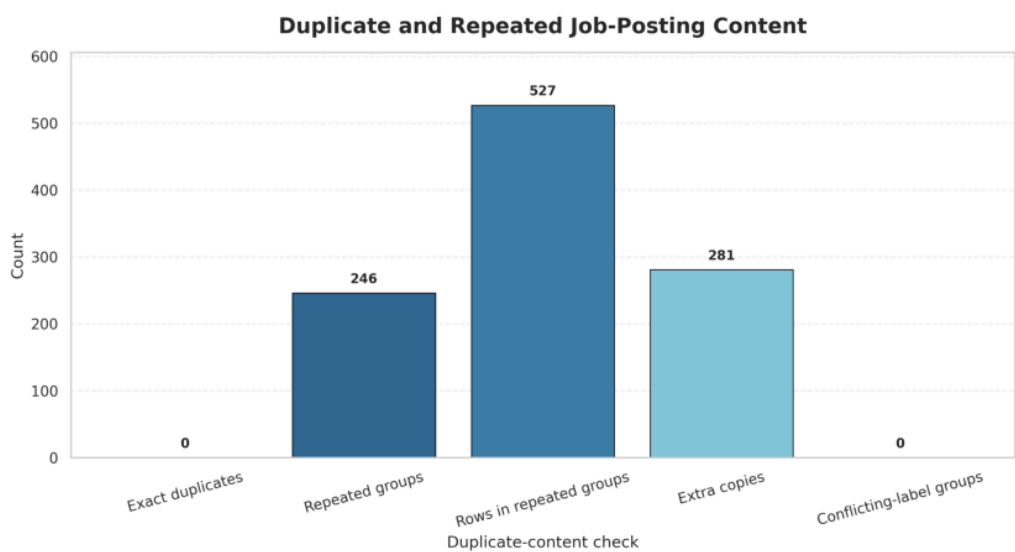


What the graph shows:
EMSCAD contains 17,880 job postings. Among them, 17,014 postings are genuine and 866 are fraudulent. Genuine postings represent 95.16% of the dataset, while fraudulent postings represent only 4.84%.

Research interpretation:
The target class is severely imbalanced. A model predicting every posting as genuine would achieve more than 95% accuracy while detecting no fraudulent jobs. Accuracy alone is therefore not a sufficient evaluation metric.

Modelling implication:
Model evaluation must focus on fraud-class precision, recall, F1-score, PR-AUC, balanced accuracy and MCC. Class weights and threshold optimisation should be performed using training and validation data only.

Figure 3: Duplicate and Repeated-Content Analysis



What the graph shows:

The duplicate-content analysis identified 246 repeated-content groups containing 527 rows. These groups contain 281 additional copies beyond the first occurrence. No repeated-content group was found with conflicting genuine and fraudulent labels.

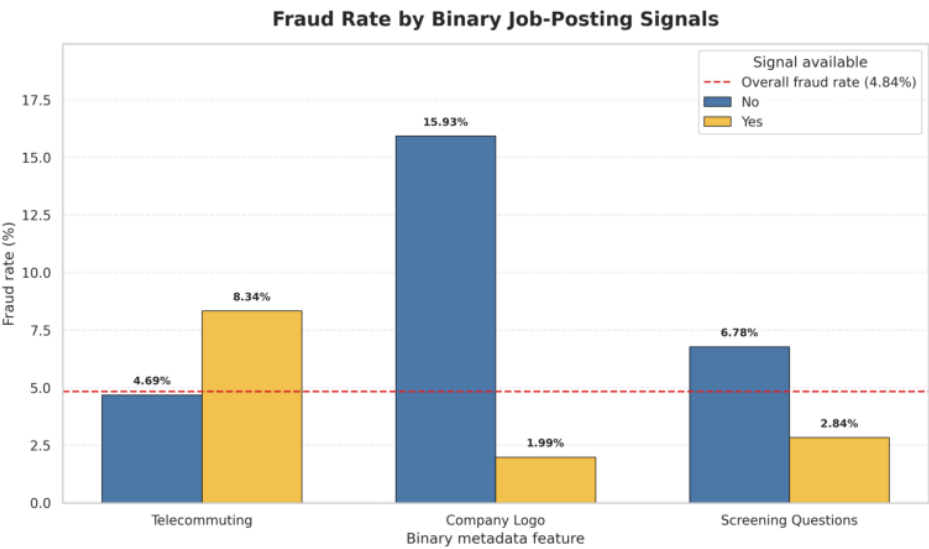
Research interpretation:

Although exact duplicate rows are absent when job_id is included, many postings reuse the same content with different identifiers. A random row-level split could place nearly identical postings in both training and evaluation sets.

Modelling implication:

Repeated postings must be assigned to campaign groups before data splitting. All rows belonging to one campaign should remain within only one of the training, validation or test partitions.

Figure 4: Fraud Rates of Binary Metadata Features



What the graph shows:

The graph compares fraud rates for telecommuting, company-logo availability and screening questions. Telecommuting postings show an 8.34% fraud rate compared with 4.69% for non-telecommuting postings. Jobs without a company logo show a 15.93% fraud rate, compared with 1.99% when a logo is available. Jobs without screening questions show a 6.78% fraud rate, compared with 2.84% when questions are present.

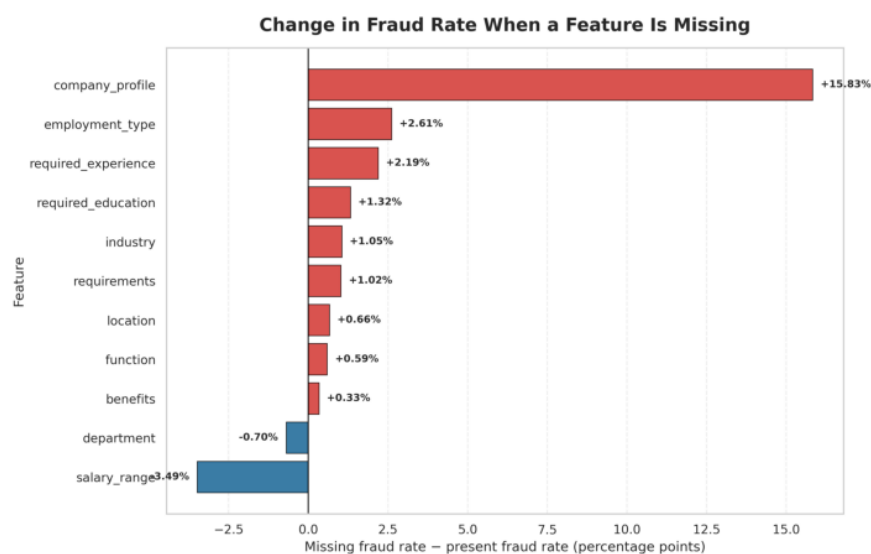
Research interpretation:

The absence of a company logo is the strongest binary metadata signal. However, these values represent associations and should not be treated as independent rules for declaring a posting fake.

Modelling implication:

The three binary fields should be retained as metadata features and combined with textual, semantic and campaign evidence.

Figure 5: Missingness as a Fraud Signal



What the graph shows:

This graph compares the fraud rate when each feature is missing against its fraud rate when present. Missing company profiles show a fraud rate of approximately 17.74%, compared with 1.91% when present. Missing employment type and experience also show higher fraud rates. Salary-range missingness behaves differently: missing salary has a lower fraud rate than available salary information.

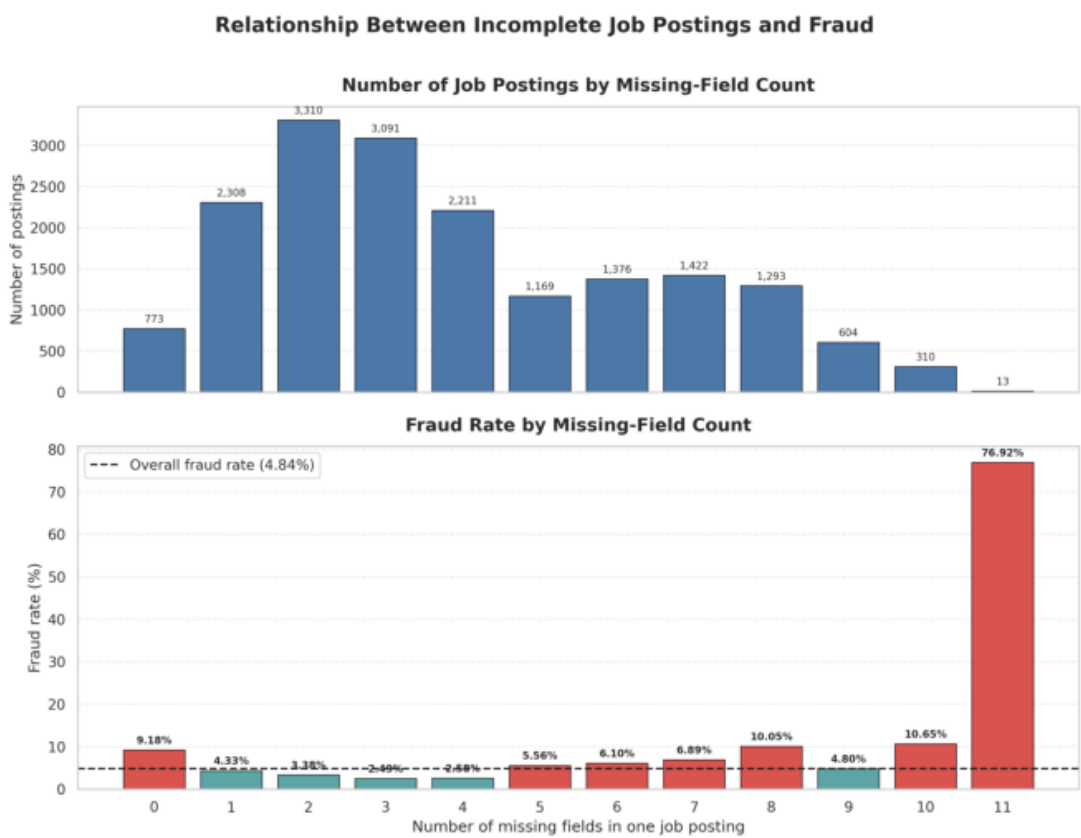
Research interpretation:

Missingness has feature-specific meaning. It cannot be handled using one universal rule. The missing-description result is based on only one record and must not be interpreted as a reliable 100% fraud pattern.

Modelling implication:

Create one missingness flag per field. The model should learn the importance of each flag independently instead of using only a single total-missing-values feature.

Figure 6: Number of Missing Fields versus Fraud Rate



What the graph shows:

The graph studies whether fraud increases as more fields become missing. Fraud rates do not increase in a perfectly linear manner. Posts with eight or ten missing fields show fraud rates around 10%, while the very high rate for eleven missing fields is based on only thirteen postings.

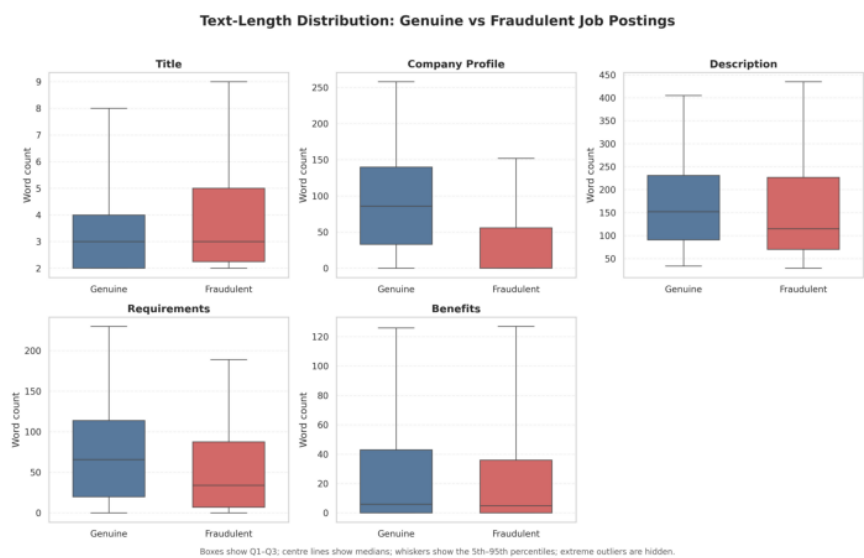
Research interpretation:

The total number of missing fields provides useful information, but extreme groups with very few observations can create unstable percentages. Individual missing fields remain more informative than the total count alone.

Modelling implication:

Retain both the total missing-field count and field-level missing indicators. Tree-based models can learn nonlinear interactions between these variables.

Figure 7: Text-Length Distribution by Class



What the graph shows:

The box plots compare title, company-profile, description, requirements and benefits lengths between genuine and fraudulent postings. Fraudulent titles are slightly longer on average. Fraudulent descriptions and requirements are generally shorter, and fraudulent company profiles contain fewer words than genuine company profiles.

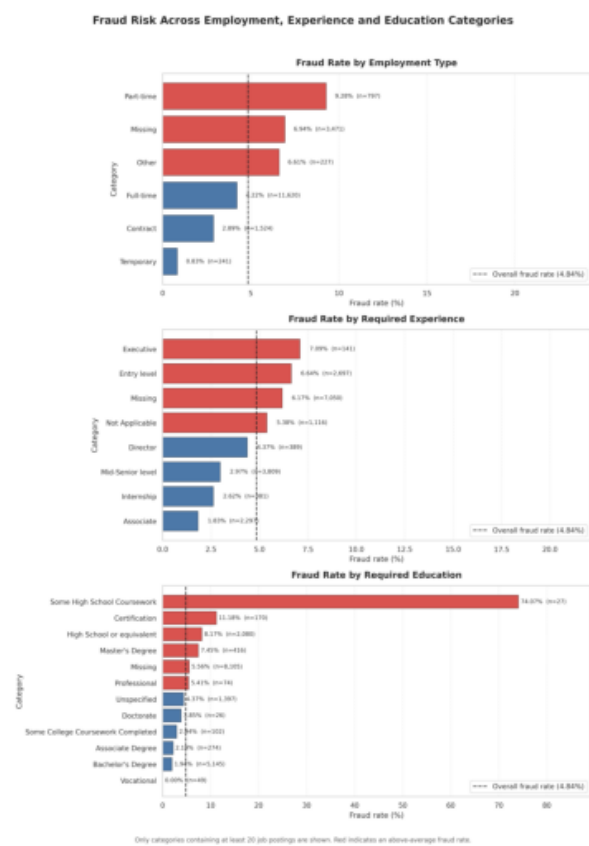
Research interpretation:

Fraudulent postings differ not only in the words used but also in how information is structured. The distributions overlap, meaning text length cannot independently separate the two classes.

Modelling implication:

Word count, character count, sentence count and section-presence features should be included in the metadata branch while the full text is analysed by lexical and semantic branches.

Figure 8: Categorical Feature Fraud Rates



What the graph shows:

This figure compares fraud rates across employment type, required experience and required education categories. Part-time jobs show a higher fraud rate than full-time and temporary positions. Entry level and executive categories show elevated risk in comparison with several other experience levels. Some education categories show high percentages but contain very few observations.

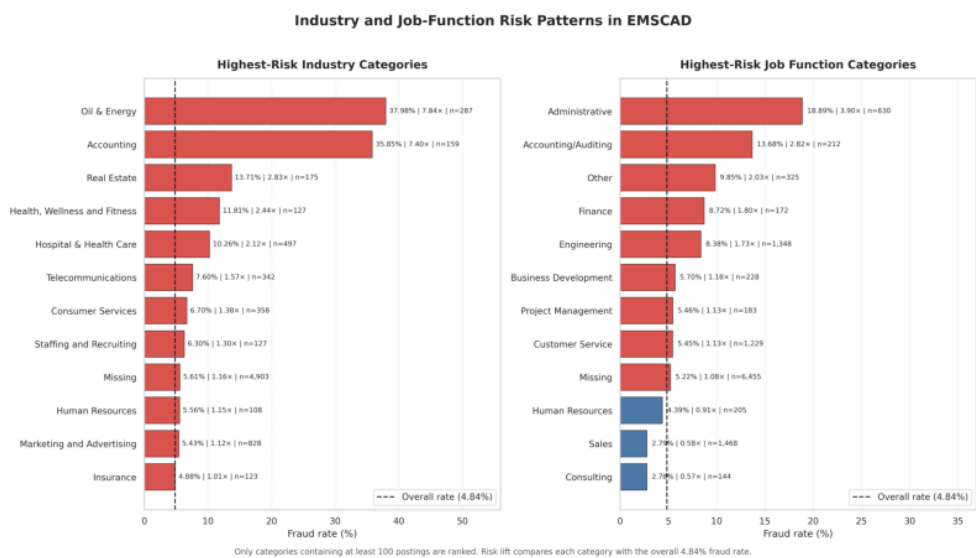
Research interpretation:

Categorical features contain useful risk information, but rare categories may produce misleadingly high rates. These categories should not be converted into manual fraud rules.

Modelling implication:

Use a model capable of handling categorical features and nonlinear interactions. Category frequency and sample size must be considered when interpreting feature importance.

Figure 9: Industry and Job-Function Fraud Risk

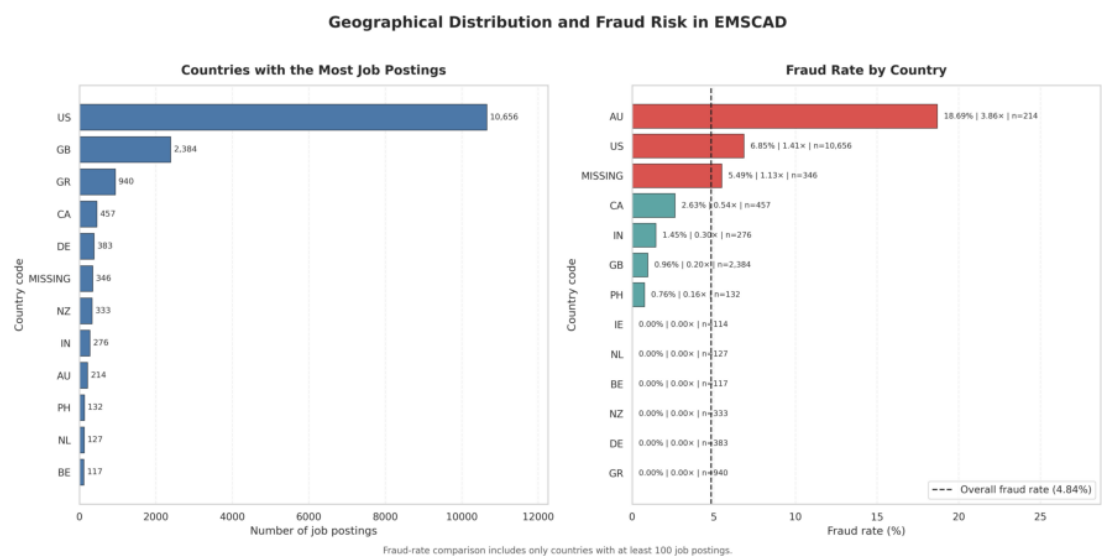


What the graph shows:
Oil and Energy, Accounting and Real Estate show high observed fraud rates. Administrative and Accounting/Auditing functions also show elevated risk. Other categories have lower fraud rates or contain fewer fraudulent examples.

Research interpretation:
Industry and function are useful contextual signals. However, some high-risk categories may be connected to a small number of repeated scam campaigns rather than representing universal industry risk.

Modelling implication:
Industry and function should be used with campaign isolation and appropriate categorical encoding. They should never be used to automatically reject all postings from one industry.

Figure 10: Country Distribution and Fraud Risk



What the graph shows:

The United States contains the largest number of postings and an observed fraud rate of approximately 6.85%. Australia shows a higher rate of approximately 18.69%, while Canada, India and the United Kingdom show lower rates. Several countries contain no fraudulent examples in EMSCAD.

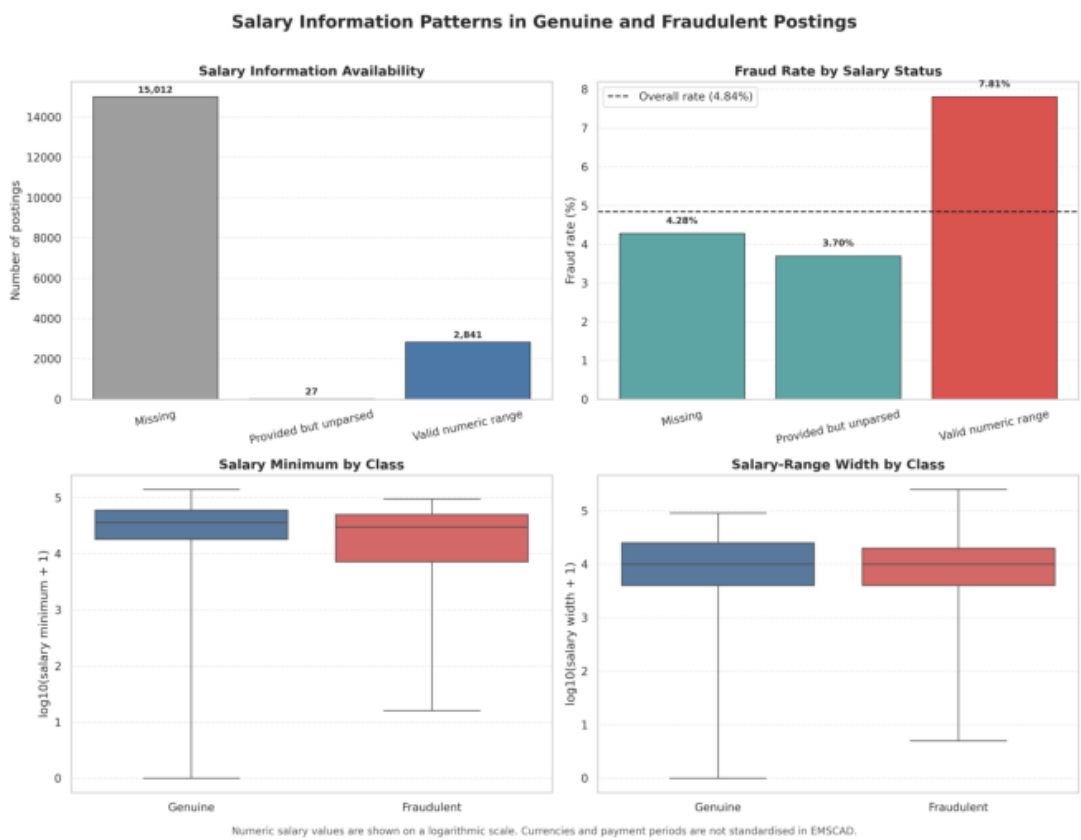
Research interpretation:

Country-level fraud rates are strongly affected by the number of available records and the campaigns represented in the dataset. A zero observed fraud rate does not prove that a country is free from fraudulent job postings.

Modelling implication:

Extract country from location and use it as supporting metadata. Rare and unseen countries require safe handling during real-world inference.

Figure 11: Salary-Range Availability and Fraud Risk



What the graph shows:

Most postings do not provide salary information. Missing salary ranges show an observed fraud rate of approximately 4.28%. Valid numeric salary ranges show a higher fraud rate of approximately 7.81%, while provided but unparsed values form a very small group.

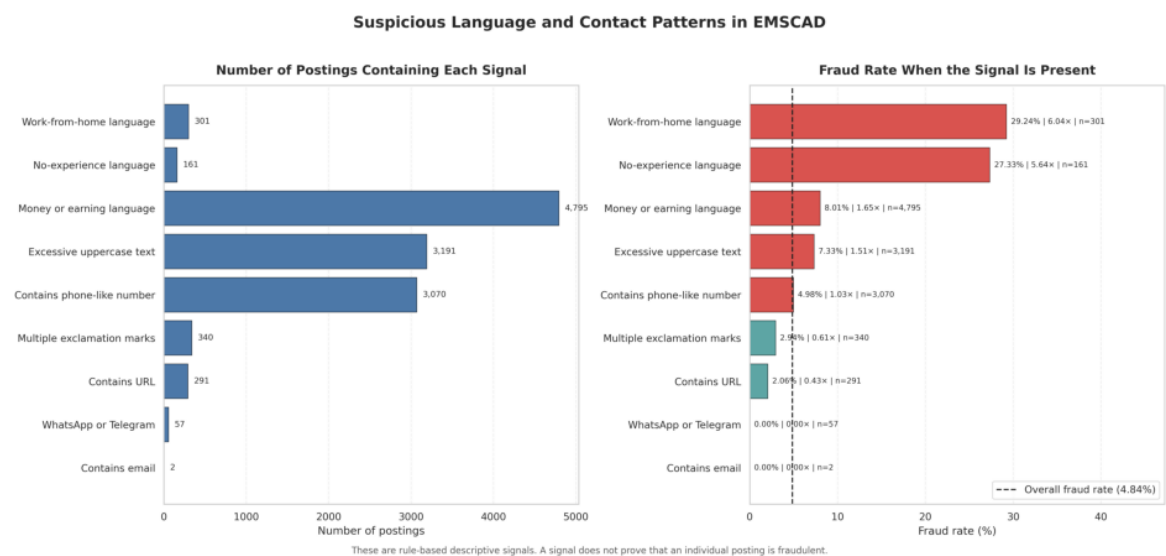
Research interpretation:

Salary availability is not automatically a sign of legitimacy. Some fraudulent postings may use attractive salary values as part of their recruitment message.

Modelling implication:

Use salary availability, parsed minimum, parsed maximum, salary width and parsing-status features. Extreme salary values should be handled carefully rather than deleted automatically.

Figure 12: Suspicious Language Signals



What the graph shows:

Work-from-home language shows an observed fraud rate of 34.68% and a risk lift above seven. No-experience language shows a 25.90% fraud rate, while money or earning language also increases fraud risk. Excessive uppercase text has a high percentage but occurs in very few postings. Phone numbers, URLs and multiple exclamation marks are weaker standalone signals.

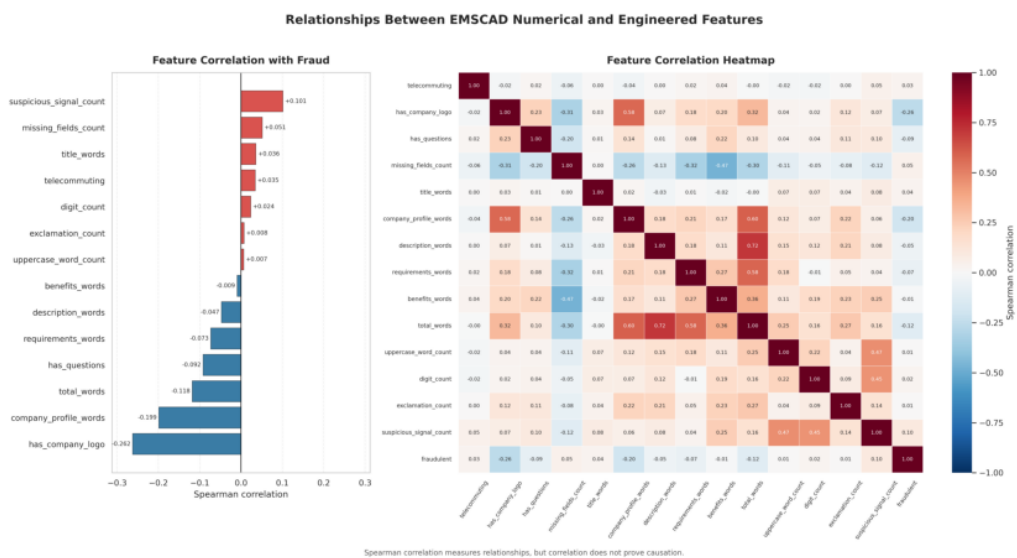
Research interpretation:

Several understandable language patterns are associated with fraud, but no single keyword can reliably classify every posting. Some apparently suspicious signals are rare or occur in genuine postings as well.

Modelling implication:

Create interpretable text-signal flags while retaining word and character TF-IDF features. These flags improve explainability and should support, not replace, learned textual representations.

Figure 13: Feature Correlation Heatmap



What the graph shows:

The heatmap presents pairwise relationships among binary, count, missingness and engineered metadata variables. Most features have weak or moderate individual relationships with fraud. Some length and missingness features are correlated with one another because they describe related properties of the same posting sections.

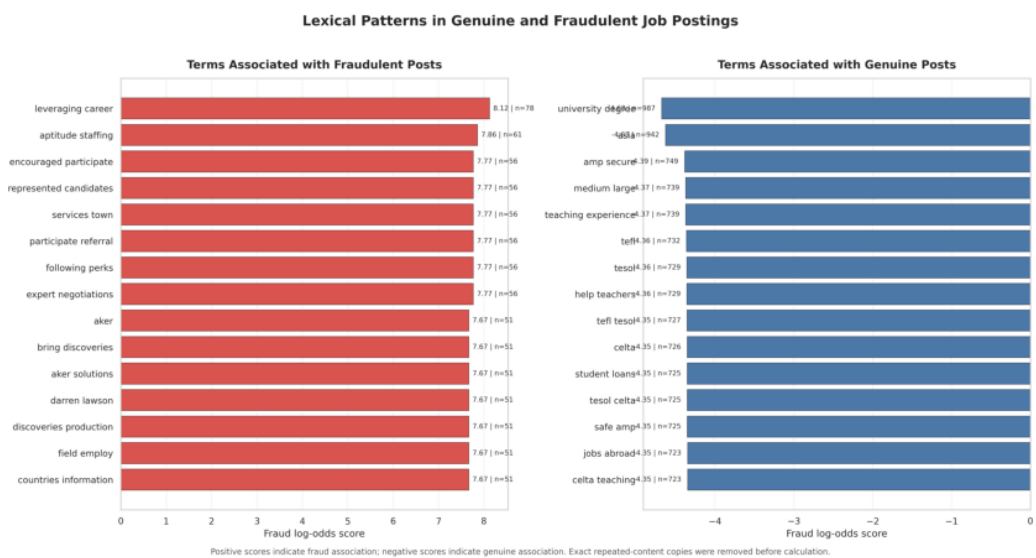
Research interpretation:

Fraud is not explained by one dominant numerical variable. Useful evidence is distributed across multiple weak signals. Correlation also does not establish causation.

Modelling implication:

Use multivariate models and regularisation. Highly related features should be monitored, while tree-based models can capture nonlinear interactions that are not visible in a correlation matrix.

Figure 14: Fraud- and Genuine-Associated Terms



What the graph shows:

This figure displays terms and phrases that appear more frequently in fraudulent or genuine postings. Some fraud-associated terms are linked to specific staffing, petroleum, administrative or referral templates. Genuine-associated terms include education, teaching and qualification-related phrases.

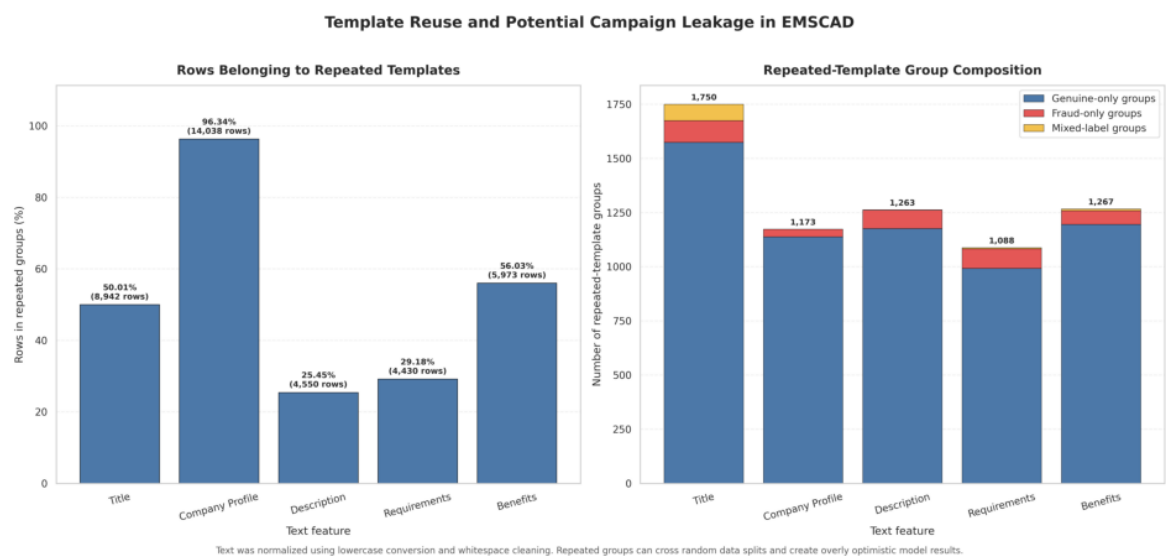
Research interpretation:

Term association helps explain lexical differences, but highly specific phrases may identify repeated campaigns or organisations instead of general fraud behaviour. Random splitting could allow the model to memorise these phrases.

Modelling implication:

Use word and character n-grams with regularisation and campaign-isolated evaluation. Term importance should be presented as an association rather than proof of fraud.

Figure 15: Repeated Template Analysis



What the graph shows:

Titles contain 1,810 repeated-template groups covering 9,157 rows, with the largest title group containing 406 postings. Company profiles show especially strong reuse, with repeated templates covering approximately 14,040 rows and a largest group of 726. Descriptions, requirements and benefits also contain substantial template repetition.

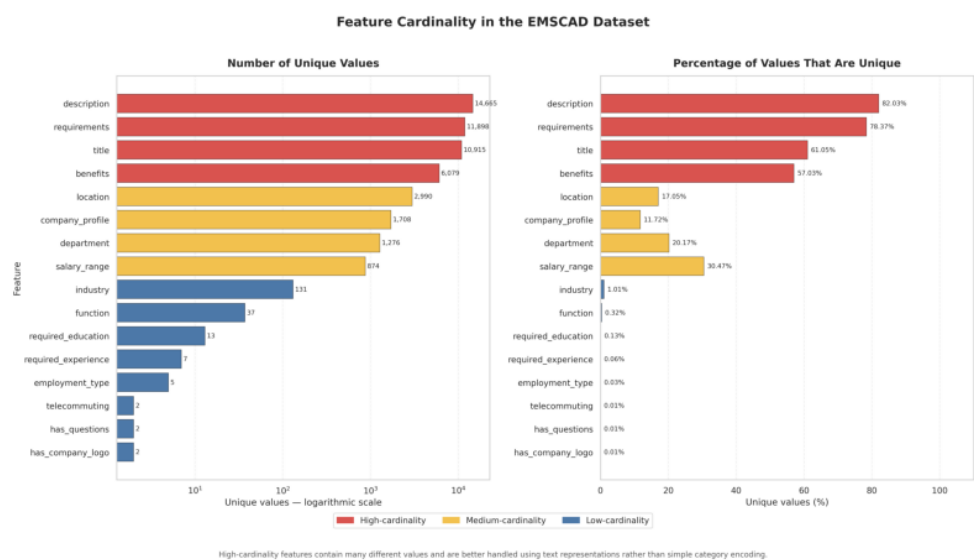
Research interpretation:

Template reuse is one of the most important properties of EMSCAD. A model evaluated using a normal random split may see nearly the same campaign wording during both training and testing, producing overly optimistic performance.

Modelling implication:

Generate a campaign_group identifier from normalized repeated templates. Use group-isolated training, validation, testing and five-fold out-of-fold evaluation.

Figure 16: Feature Cardinality



What the graph shows:

Description, requirements, title and benefits contain very high numbers of unique values. Description has approximately 14,801 unique non-missing values, requirements about 11,967, and title about 11,231. Industry, function, education, experience and employment type have much lower cardinality.

Research interpretation:

High-cardinality text fields should not be treated as ordinary categorical variables. Low-cardinality structured fields are more suitable for categorical models.

Modelling implication:

Process long text using lexical and semantic representations. Use categorical encoding or CatBoost for structured fields, with an explicit strategy for rare and unseen categories.

Categorical Feature Significance Tests

Feature	Levels	After_Rare_Groups	Chi_Square	Degrees_of_Freedom	P_Value	Cramers_V	Effect_Size	Adjusted_P_Value_Bonferroni
industry	75		1543.8100	74	0.0000	0.2867	Small	0.0000
has_company_logo	2		1224.0637	1	0.0000	0.2615	Small	0.0000
country	36		569.5471	35	0.0000	0.1729	Small	0.0000
function	37		546.7466	36	0.0000	0.1690	Small	0.0000
required_education	13		463.8083	12	0.0000	0.1590	Small	0.0000
has_questions	2		149.2600	1	0.0000	0.0911	Negligible	0.0000
required_experience	8		126.6599	7	0.0000	0.0818	Negligible	0.0000
employment_type	6		99.8286	5	0.0000	0.0728	Negligible	0.0000
salary_status	3		64.6985	2	0.0000	0.0592	Negligible	0.0000
telecommuting	2		20.5237	1	0.0000	0.0330	Negligible	0.0000

The table reports statistical association tests for categorical features. Statistical significance does not automatically imply high predictive importance or causation.

Continuous Feature Significance Tests

	Feature	Genuine_Median	Fraud_Median	Mann_Whitney_U	P_Value	Rank_Biserial_Correla	Effect_Size	Adjusted_P_Value_B
company_profile_word_c		88.0000	0.0000	3426365.0000	0.0000	-0.5349	Large	0.0000
	total_words	372.0000	241.0000	5015213.0000	0.0000	-0.3192	Medium	0.0000
suspicious_signal_cou		0.0000	0.0000	9130093.0000	0.0000	0.2393	Small	0.0000
requirements_word_cou		67.0000	35.0000	5908652.0000	0.0000	-0.1980	Small	0.0000
	missing_fields_count	3.0000	5.0000	8374904.0000	0.0000	0.1368	Small	0.0000
description_word_cou		154.0000	119.0000	6424672.0000	0.0000	-0.1279	Small	0.0000
	title_word_count	3.0000	3.0000	7991807.5000	0.0000	0.0848	Negligible	0.0000
	digit_count	13.0000	23.0000	7835042.5000	0.0016	0.0635	Negligible	0.0022
	benefits_word_count	6.0000	5.0000	7182258.5000	0.1968	-0.0251	Negligible	0.2406
uppercase_word_cou		3.0000	3.0000	7494889.5000	0.3844	0.0174	Negligible	0.3844
	exclamation_count	0.0000	0.0000	7488375.5000	0.3135	0.0165	Negligible	0.3449

Continuous-feature tests compare genuine and fraudulent distributions. Effect size and practical usefulness should be considered together with p-values.

Final EDA Summary

Metric	Value
Rows	17880.0000
Columns	18.0000
Genuine posts	17014.0000
Fraudulent posts	866.0000
Fraud rate (%)	4.8434
Genuine-to-fraud ratio	19.6467
Exact duplicate rows	0.0000
Duplicate copies excluding j	281.0000
Categorical tests	10.0000
Continuous tests	11.0000

This table consolidates the principal EDA findings that guide preprocessing, feature engineering and evaluation.

Final Missingness Summary

Feature	Missing_Count	Missing_Percentage
salary_range	15012	83.9597
department	11553	64.6141
required_education	8105	45.3300
benefits	7220	40.3803
required_experience	7050	39.4295
function	6455	36.1018
industry	4903	27.4217
employment_type	3471	19.4128
company_profile	3308	18.5011
requirements	2698	15.0895
location	346	1.9351
description	2	0.0112
title	0	0.0000
has_questions	0	0.0000
telecommuting	0	0.0000
has_company_logo	0	0.0000

Missingness is preserved using explicit categories and indicators instead of deleting incomplete rows.

EDA-Based Modelling Recommendations

1. Leakage-safe splitting

Construct campaign groups from repeated normalized templates. Keep every campaign inside only one partition. Use five-fold group-isolated out-of-fold predictions for calibration and fusion training.

2. Text representation

Combine title, company profile, description, requirements and benefits using explicit section markers. Use both word and character TF-IDF features and section-aware semantic embeddings.

3. Metadata representation

Retain binary indicators, categorical fields, text lengths, missingness flags, country, salary parsing information and suspicious-language counts.

4. Class imbalance

Use class weighting and validation-based threshold selection. Do not judge the model using accuracy alone.

5. Evaluation metrics

Report precision, recall, F1-score, specificity, balanced accuracy, PR-AUC, ROC-AUC and MCC, together with confusion matrix counts.

6. Real-world use

Use the final score as a decision-support signal for manual review. Do not automatically delete or block a posting based only on one feature or keyword.

EDA Limitations

Dataset imbalance

Fraudulent examples represent only 4.84% of EMSCAD. Results for very small subgroups can be unstable.

Campaign concentration

Many fraudulent postings belong to repeated campaigns. Performance on familiar templates may not represent performance on completely unseen scams.

Geographical representation

The country distribution is unequal. Country-level fraud rates should not be interpreted as universal geographic risk.

Term interpretation

Fraud-associated terms may identify specific organisations or templates rather than general scam language.

External validity

An untouched campaign-isolated test is harder and more realistic than a normal random split, but additional modern real-world labelled data is still required.

EDA Conclusion

Overall conclusion

The EMSCAD dataset contains strong lexical, metadata, missingness and campaign-reuse signals for fraudulent job posting detection. However, severe class imbalance and repeated templates make normal accuracy and random splitting unsafe for research conclusions.

Architecture decision

The EDA supports a four-view model containing lexical, metadata, semantic and campaign branches. These complementary views can capture wording patterns, structural behaviour, section meaning and template novelty.

Evaluation decision

Campaign-group isolation, locked validation, a frozen decision threshold and one-time untouched testing are required to produce a credible estimate of model generalisation.

Research relevance

These findings provide the evidence-based foundation for the SAGE-FJD reliability-aware multimodal architecture and its human-review deployment policy.

Appendix: EDA Source Tables

Available reproducibility tables

1. 17_categorical_significance_tests.csv
2. 18_continuous_significance_tests.csv
- 3.
- 19_final_eda_summary.csv
4. 20_final_missingness_summary.csv
5. all_term_association_scores.csv
- 6.
- binary_metadata_fraud_rates.csv
7. categorical_feature_fraud_rates.csv
8. class_distribution.csv
- 9.
- country_fraud_risk.csv
10. duplicate_content_summary.csv
11. feature_cardinality_summary.csv
- 12.
- feature_correlations_with_fraud.csv
13. industry_function_fraud_risk.csv
- 14.
- missing_fields_per_row_fraud_rate.csv
15. missing_values_summary.csv
16. missingness_fraud_signal.csv
- 17.
- repeated_content_group_sizes.csv
18. repeated_template_summary.csv
19. salary_status_fraud_rate.csv
- 20.
- spearman_correlation_matrix.csv
21. suspicious_text_signal_flags.csv
- 22.
- suspicious_text_signal_summary.csv
23. text_length_row_data.csv
24. text_length_summary_by_class.csv
- 25.
- top_fraud_associated_terms.csv
26. top_genuine_associated_terms.csv
27. valid_salary_summary_by_class.csv